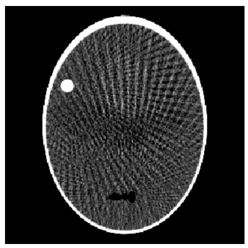
CT Reconstruction Bench

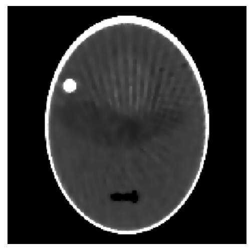
User Manual and Physics Reference

An interactive, browser-based simulator of the complete computed-tomography imaging chain: polychromatic X-ray source, energy-integrating or photon-counting detector, artifact generation, analytic and iterative reconstruction, post-processing denoisers, and quantitative image-quality measurement. Everything runs client-side in a single HTML file with no network access and no external libraries.

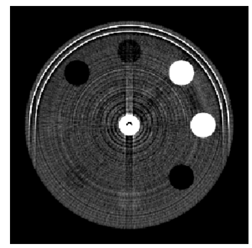
Head phantom



FBP, 45 views



SIRT + TV, 45 views



Ring artifact

Application file	ct_recon_bench.html (single self-contained file)
Manual revision	1.0 — 28 August 2026
Requirements	Any modern desktop browser. No installation, no server, no internet connection, no stored data.
Intended use	Teaching, parameter-trade-off demonstration and design intuition. Not a validated clinical or regulatory tool.

How to use this manual

Section 2 explains how a single result is produced, stage by stage, and is worth reading once before touching the controls. Section 4 is the reference for every control on the page and can be consulted out of order. Sections 5 to 7 give the mathematics behind the reconstruction methods, the denoisers and the metrics. Section 9 contains guided experiments with the numbers you should expect to see, which is the fastest way to confirm the tool is behaving.

1. Introduction

1.1 What the tool does

The bench takes a numerical phantom, simulates a fan of X-ray measurements through it, corrupts those measurements with the physical effects that limit real CT systems, reconstructs an image from them, and then measures how good that image is. Every stage is exposed as a control, so the cost of any single design choice — a smaller voxel, a smoother kernel, half the dose, a photon-counting sensor instead of a scintillator — can be read directly off the metric panel rather than argued about.

Nothing in the image chain is faked. Beam hardening is not painted onto the image as a shading term; it emerges because the transmitted signal is summed over five energy bins with energy-dependent attenuation coefficients before the logarithm is taken. Noise is not additive Gaussian texture; it is Poisson sampling of transmitted quanta, propagated through the log and the reconstruction. Streaks near dense implants appear because the rays through them genuinely run out of photons.

1.2 Opening the application

Open `ct_recon_bench.html` in a browser, by double-clicking it or dragging it onto a browser window. It runs entirely on the local machine. Nothing is uploaded, nothing is stored, and no external resources are fetched, so it works offline and inside restricted networks.

1.3 Screen layout

Region	Contents
`Left column`	All controls, grouped into Scenario, Acquisition geometry, Detector & dose, Beam & artifacts, Reconstruction, Denoising and Display. Changing any control re-runs the simulation.
`Image row`	Ground truth (the phantom at 70 keV), sinogram (the measured data), reconstruction, and a signed error map of reconstruction minus truth on a blue-to-red scale at plus or minus 250 HU.
`Plot row`	Central horizontal profile through truth and reconstruction; modulation transfer function with the voxel Nyquist limit marked; radial noise power spectrum.
`Metric panel`	Twelve quantitative readouts, colour-coded green, amber or red against fixed acceptance thresholds.
`Pin bar`	Pin stores the current result as a thumbnail card with its settings and key metrics, so a changed parameter can be compared against the previous state side by side.

1.4 Conventions used throughout

Attenuation is reported as the linear attenuation coefficient in inverse centimetres, written mu. Images are displayed in Hounsfield units, defined against the reference water value at 70 keV. Spatial frequency is given in line pairs per millimetre. Noise is the standard deviation of Hounsfield values inside a uniform region of interest, written SD. Angles are measured over a 180 degree parallel-beam rotation.

$$\mathrm{HU} = 1000 \times (\mathrm{mu} - \mathrm{mu}_{water}) / \mathrm{mu}_{water} , \quad \mathrm{mu}_{water} = 0.195\ \mathrm{cm}^{-1} \text{ at } 70\ \mathrm{keV}$$

2. How a result is generated

Each time a control changes, the following seven stages run in order. The progress bar under the controls names the stage currently executing.

Stage	Operation	Output
`1. Phantom`	The selected scenario is expanded into a list of ellipses, each carrying a material composition vector.	Object list
`2. Projection`	Exact analytic line integrals through every ellipse, accumulated separately for each material, for every view and detector channel.	Path lengths per material
`3. Spectrum`	Path lengths are converted to transmitted quanta in each of five energy bins using the tabulated attenuation coefficients.	Expected counts
`4. Detection`	Poisson sampling, energy weighting or unity weighting, charge sharing, electronic noise, channel gain errors.	Measured signal
`5. Corruption`	Scatter field added; the logarithm is taken against the air reference to give line integrals.	Sinogram
`6. Reconstruction`	Filtered backprojection, unfiltered backprojection, SIRT, or SIRT with total-variation regularisation.	Image in mu
`7. Analysis`	Conversion to Hounsfield units, optional denoising, then region-of-interest statistics, MTF, NPS and error metrics.	Image and metrics

2.1 Stage 1 — the phantom

Phantoms are built as sums of ellipses rather than as pixel maps. Each ellipse carries a centre, two semi-axes, a rotation, and a composition dictionary that assigns weights to materials, for example a vessel described as eighty per cent soft tissue and twenty per cent iodinated blood. Weights may be negative, which is how an interior structure replaces the material of the body it sits inside without double counting.

Because the phantom is analytic, the ground truth is exact at any matrix size, and the forward projection is not limited by the resolution of a discretised object. This matters: if the phantom were a pixel grid, the projection would inherit the grid's aliasing and the measured resolution would be an artefact of the simulation rather than of the imaging chain.

2.2 Stage 2 — analytic line integrals

For a parallel-beam geometry, a ray is specified by its angle and its signed distance from the isocentre. For an ellipse of semi-axes a and b centred at the origin, the chord length along that ray has a closed form. Every ellipse is transformed into its own local frame, the chord is evaluated, and the result is accumulated into the path-length array for each material in that ellipse's composition.

$$L_m(\text{theta}, s) = \text{sum over ellipses of } w_m \times \text{chord}(\text{theta}, s)$$

The array is indexed by material, view and channel, so the spectral stage can weight each material by its own energy-dependent attenuation without re-tracing any rays. When aperture integration is enabled, three rays are traced across the width of each detector element and averaged, which reproduces the resolution loss caused by finite element size.

2.3 Stage 3 — the X-ray spectrum and attenuation table

The beam is represented by five energy bins with relative photon fluence weights chosen to approximate a filtered 120 kVp tungsten spectrum. The mean energy of this discrete spectrum is 74.8 keV.

Energy bin	40 keV	60 keV	80 keV	100 keV	120 keV
Relative fluence	0.12	0.30	0.28	0.20	0.10

Linear attenuation coefficients in inverse centimetres are tabulated per material and per bin. The monochromatic reference used for the ground truth and for the Hounsfield conversion is 70 keV, evaluated as the mean of the 60 and 80 keV entries.

Material	40 keV	60 keV	80 keV	100 keV	120 keV	70 keV ref	HU
Air	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	-1000
Lung	0.0670	0.0515	0.0460	0.0428	0.0403	0.0488	-750
Fat	0.2270	0.1830	0.1670	0.1570	0.1490	0.1750	-103
Soft tissue, low	0.2626	0.2019	0.1803	0.1676	0.1578	0.1911	-20
Soft tissue (water)	0.2680	0.2060	0.1840	0.1710	0.1610	0.1950	0
Soft tissue, high	0.2734	0.2101	0.1877	0.1744	0.1642	0.1989	+20
Muscle	0.2760	0.2124	0.1897	0.1763	0.1660	0.2011	+31
Cortical bone	1.2780	0.6040	0.4280	0.3560	0.3200	0.5160	+1646
Iodinated blood	2.0000	0.9500	0.5800	0.4400	0.3600	0.7650	+2923
Implant alloy	5.4700	1.8300	1.0300	0.7500	0.6200	1.4300	+6333

Linear attenuation coefficient in cm^{-1}. The three soft-tissue entries differ by twenty Hounsfield units and are used to build low-contrast detail.

The energy dependence is what produces beam hardening. As the beam crosses the object the low energy bins are removed preferentially, the surviving beam attenuates less per centimetre, and the logarithm of the total transmitted signal is no longer proportional to path length. The reconstruction consequently under-reports mu in the centre of a large object, which appears as cupping and as dark bands between dense structures.

2.4 Stage 4 — the detector

Expected transmitted quanta in bin e for channel i follow Beer-Lambert attenuation of the incident fluence for that bin. Those expectations are sampled with a Poisson generator seeded from the seed control, so a given seed reproduces a given noise realisation exactly.

$$n_{e,i} \sim \mathrm{Poisson}[\ N_0\ w_e\ g_i\ \exp(\ -\ \mathrm{sum}_m\ \mathrm{mu}_m(E_e)\ L_{m,i}\ )\ ]$$

The two detector models then differ in how those quanta are converted into a signal. An energy-integrating detector weights each quantum by its energy, which over-weights the high energy photons that carry the least contrast, and adds a Gaussian electronic noise term that is independent of signal level. A photon-counting detector weights every quantum equally and adds no electronic term, but can miscount through charge sharing, modelled here as a fraction of each channel's signal redistributed to its two neighbours.

$$\mathrm{EID:}\ S_i = \mathrm{sum}_e\ E_e\ n_{e,i} + \mathrm{sigma}_{el} \times E_{mean} \times N(0,1)\quad \mathrm{PCD:}\ S_i = \mathrm{sum}_e\ n_{e,i}$$

The practical consequence is visible in the metric panel. At high fluence the two models give similar noise, because both are quantum limited. As dose falls, the electronic term becomes comparable to the signal in the most attenuated channels and the energy-integrating detector degrades much faster. This is the quantitative form of the argument for direct-conversion sensors at low dose, and the bench reproduces it without any special-case code.

2.5 Stage 5 — scatter and the logarithm

Scatter is modelled as a broad, low-frequency field obtained by smoothing the noise-free primary signal with a Gaussian whose width is one eighth of the detector array, then scaling that field so that the mean scatter-to-primary ratio inside the object shadow equals the requested value. Adding a positive, slowly varying term before the logarithm suppresses the apparent attenuation most where the primary is weakest, which is why uncorrected scatter produces broad shading, a depressed Hounsfield level and dark streaks between dense structures.

$$p_i = -\ln[(S_i + k \times scatter_i) / S_0]$$

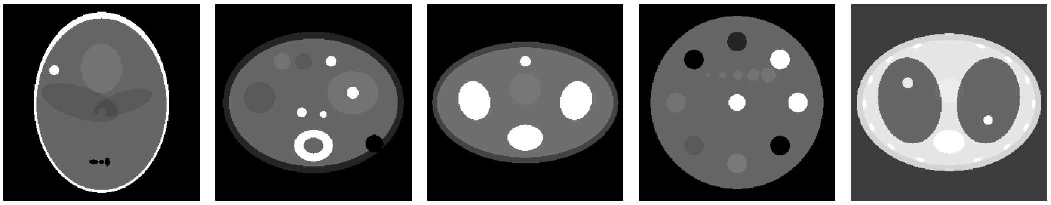
The reference signal S_0 is the unattenuated air signal computed with the same spectrum and the same detector weighting, so the line integrals are correctly normalised for both detector models. If a channel receives essentially no signal the logarithm is clamped, which is the numerical counterpart of photon starvation.

2.6 Stages 6 and 7

Reconstruction is covered in Section 5, post-processing in Section 6 and measurement in Section 7. One implementation detail is worth stating here because it governs whether the iterative methods converge: the projector used inside SIRT and its backprojector form a matched adjoint pair. The forward operator is ray-driven and the backprojector deposits each ray's residual back onto the same voxels with the same weights. The inner-product identity was verified numerically to six decimal places. An unmatched pair, which is easy to write by accident, causes the iteration to drift or diverge no matter how the relaxation is tuned.

3. Scenarios

The scenario sets the phantom, the field of view and the positions of the measurement regions. Changing scenario resets nothing else, so a comparison can be carried across phantoms.



Head **Abdomen** **Pelvis** **QA phantom** **Thorax**

Ground truth for the five scenarios, rendered at 70 keV.

Scenario	FOV	Contents and purpose
`Head — Shepp-Logan`	`220 mm`	The classical test object: a dense skull ring, ventricle-like structures and small low-contrast inserts. Thick bone around a small object makes this the natural scenario for beam hardening and for sparse-view work.
`Abdomen — iodine and low contrast`	`380 mm`	A large body with a subcutaneous fat ring, vertebral bone, iodinated vessels and paired lesions at plus and minus twenty Hounsfield units. The natural scenario for dose and contrast-detail trade-offs.
`Pelvis — bilateral implants`	`340 mm`	Two femoral heads with alloy stems. Rays passing through both implants lose almost all their photons, producing streaks and dark bands. The natural scenario for iterative reconstruction.
`QA phantom`	`250 mm`	A uniform water cylinder with eight material rods on a circle, a low-contrast ladder and a central bone rod. Uniform background and clean edges make this the scenario for measuring rather than for looking.
`Thorax — lung and moving nodule`	`400 mm`	Aerated lungs, a rib cage and two nodules flagged as moving. The natural scenario for motion artifacts and for wide-window display.

3.1 Measurement regions

Each scenario defines four regions of interest, drawn on the reconstruction when *Show measurement ROIs* is enabled. The background region sits in uniform material and supplies the noise standard deviation and the Hounsfield accuracy. The signal region sits on a known contrast feature and supplies the contrast-to-noise ratio. Two further regions, one central and one peripheral, supply the cupping figure. A separate rod defines the edge used for the modulation transfer function.

Every region was checked against the analytic phantom and contains exactly one material, with zero standard deviation in the ground truth. This matters more than it sounds: a background region overlapping a rod by even a few voxels inflates the reported noise by an order of magnitude and makes every comparison meaningless.

4. Parameter reference

Every control is listed below with its range, its default and what it changes. Defaults are chosen to give a clinically plausible head scan at moderate dose.

4.1 Acquisition geometry

Control	Values	Default	Effect
`Reconstruction matrix`	64, 128, 256, 512	256	Number of voxels across the field of view. The readout converts this to millimetres using the scenario FOV, so at 220 mm the four settings give 3.44, 1.72, 0.86 and 0.43 mm voxels. Smaller voxels raise the sampling limit on resolution but collect fewer quanta each, so noise rises roughly as the inverse of voxel area for fixed dose. Compute time scales with the square of this number.
`Detector elements`	128, 256, 512	256	Number of channels spanning the detector arc. Element pitch is 1.05 × FOV divided by this number, giving 1.80, 0.90 and 0.45 mm at the head FOV. This is the hardware limit on resolution: no reconstruction can recover detail finer than the detector samples, which is why increasing the matrix alone stops helping once the detector is the bottleneck.
`Projections over 180 degrees`	45, 90, 180, 360, 720	360	Number of angular samples. Below roughly the number of detector channels multiplied by pi divided by two, the data are angularly undersampled and filtered backprojection produces the characteristic radiating streaks. Iterative methods tolerate this far better, which is the entire argument for them in sparse or interrupted acquisitions.
`Integrate over detector aperture`	on / off	on	When on, three rays are averaged across each element, so finite element width blurs the measurement as it does on real hardware. When off, each channel is a mathematical line and the measured MTF is limited only by the reconstruction, which is useful for isolating where resolution is actually being lost.

4.2 Detector and dose

Control	Range	Default	Effect
`Detector model`	EID / PCD	EID	Energy-integrating, representing an indirect-conversion scintillator, or photon counting, representing a direct-conversion sensor. The switch changes the quantum weighting, enables or disables the electronic noise floor, and enables or disables charge sharing.
`Incident quanta per element`	$10^{2.6}$ to $10^{6.5}$	$10^{5.5}$	Photon fluence per channel before attenuation, the dose proxy. Noise standard deviation scales as the inverse square root of this quantity in the quantum-limited regime. Halving dose therefore costs a factor of about 1.41 in noise, which is the exchange every protocol decision comes down to.
`Electronic noise`	0 to 120	30	Standard deviation of the additive readout noise for the energy-integrating model, in quanta-equivalent units. Independent of signal, so it dominates in highly attenuated rays and is the mechanism behind low-dose streak breakdown. Ignored for the photon-counting model.
`Charge sharing`	0 to 0.40	0.12	Fraction of each channel's signal redistributed equally to its two neighbours in the photon-counting model. Represents lateral charge diffusion and fluorescence escape in the sensor. Blurs the measurement slightly and lowers the measured MTF, which is the cost side of the photon-counting ledger. Ignored for the energy-integrating model.

4.3 Beam and artifacts

Control	Range	Default	Effect
`Polychromatic beam`	on / off	on	When on, the five-bin spectrum is used and beam hardening arises naturally. When off, the beam is monochromatic at 70 keV and the reconstruction should return the true Hounsfield values with no cupping. Toggling this is the cleanest way to separate spectral error from every other error source.
`Scatter to primary`	0 to 0.60	0	Mean ratio of scattered to primary signal inside the object shadow. Adds broad shading, depresses the overall Hounsfield level and lowers contrast. Values around 0.2 to 0.4 represent an uncorrected wide-cone acquisition.
`Detector gain non-uniformity`	0 to 3 per cent	0	Standard deviation of per-channel gain error, fixed across all views. Because the error is stationary in the detector frame it maps to concentric rings in the image, the classic signature of an imperfect flat-field calibration.
`Dead or drifting channels`	0 to 8	0	Number of channels assigned a badly wrong gain. Each produces a strong single ring and, if severe, a bright or dark arc.
`Motion amplitude`	0 to 12 mm	0	Displacement amplitude of objects flagged as moving, applied as a function of view angle so that different projections see the object in different places. Produces doubling, blurring and streaks tangent to the moving edge.
`Random seed`	1 to 99999	7	Seeds the noise generator, the gain errors and the dead-channel positions. Holding the seed fixed isolates the effect of a parameter change; varying it shows how much of an observed difference is just one noise realisation.

4.4 Reconstruction

Control	Range	Default	Effect
`Method`	FBP, BP, SIRT, SIRT+TV	FBP	See Section 5.
`Reconstruction kernel`	Ram-Lak, Shepp-Logan, Cosine, Hamming, Hann	Shepp-Logan	Apodisation applied to the ramp filter. See Section 5.2.
`Frequency cutoff`	0.15 to 1.00	1.00	Fraction of the detector Nyquist frequency above which the filter is set to zero. Lowering it truncates the ramp, which suppresses high-frequency noise and softens edges. Aggressive values introduce ringing near sharp boundaries.
`Iterations`	1 to 40	12	Number of SIRT updates. More iterations recover more detail and more noise; the iteration count is itself a regularisation parameter. Twelve is a reasonable compromise for these geometries.
`TV weight`	0 to 1	0.30	Strength of the total-variation step interleaved between SIRT updates. Zero reduces SIRT+TV to plain SIRT. High values give the flat, plastic appearance characteristic of over-regularised iterative reconstruction and can erase genuine low-contrast detail.
`Warm start from FBP`	on / off	on	Initialises the iteration with the filtered backprojection result instead of zero, typically saving several iterations. Turn it off to watch SIRT converge from nothing.

4.5 Denoising

Control	Range	Default	Effect
`Denoiser`	None, Gaussian, Median, Bilateral, TV, NLM	None	Post-reconstruction filter applied in Hounsfield units. See Section 6.
`Strength`	0.1 to 2.5	1.0	Multiplies the internal parameter of the selected filter: the Gaussian width, the bilateral spatial and range widths, the total-variation step, or the non-local means similarity bandwidth.

4.6 Display

Control	Values	Effect
`Window preset`	Soft tissue 400/40, Brain 80/40, Lung 1500/-600, Bone 2000/500, Wide 4000/0	Sets width and level together. Display windowing changes nothing in the data; it changes what is visible. A low-contrast lesion invisible at bone window is obvious at brain window.
`Window width, window level`	40 to 4000 HU, -1000 to 1500 HU	Manual override of the preset.
`Show measurement ROIs`	on / off	Overlays the regions used for the noise, contrast, cupping and resolution figures, so the reported numbers can be traced to the places they were measured.

5. Reconstruction methods

5.1 Filtered backprojection

Filtered backprojection inverts the Radon transform in two steps: each projection is convolved with a ramp filter, then the filtered projections are smeared back across the image along their original ray paths and summed. The ramp compensates for the fact that backprojection alone over-represents low spatial frequencies, which is why unfiltered backprojection produces the familiar blurred blob.

The filter is implemented as the exact discrete ramp kernel rather than by sampling the absolute-frequency response. This matters. Sampling the continuous response and inverse transforming it introduces a bias at zero frequency, which in this bench showed up as a reconstructed water value of about minus twenty-four Hounsfield units. Using the analytic space-domain kernel, transformed once and applied in the frequency domain, brings water back to within half a unit of zero.

$$h[0] = 1 / (4 \times d^2) \ , \ h[n] = -1 / (pi^2 \ n^2 \ d^2) \text{ for odd } n \ , \ h[n] = 0 \text{ for even } n$$

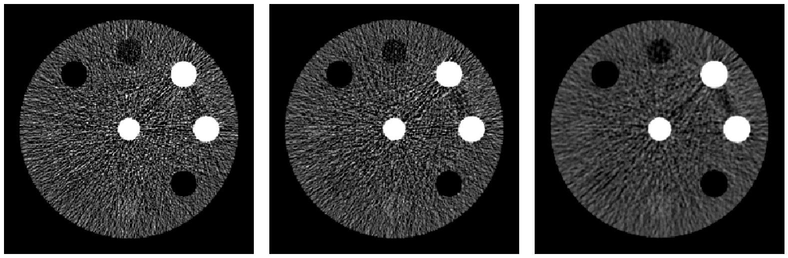
Here d is the detector sample spacing. The transformed kernel is multiplied by the chosen apodisation window and by the cutoff mask, and each projection is filtered by fast Fourier transform with zero padding to at least twice the array length to avoid circular convolution wraparound.

5.2 Reconstruction kernels

The kernel is the apodisation window multiplying the ramp. It is the single most direct control over the noise-resolution trade-off, and it costs nothing to change, which is why real scanners ship dozens of them. In the expressions below x is spatial frequency normalised to the cutoff, so x runs from zero to one and the window is zero beyond.

Kernel	Window W(x)	Character	Noise SD	MTF50
`Ram-Lak`	`1`	The unmodified ramp. Sharpest possible edges, maximum high-frequency noise. Used when resolution matters more than noise, for example bone detail.	42 HU	0.49
`Shepp-Logan`	`sin(pi x / 2) / (pi x / 2)`	Mild roll-off, the general-purpose default. Gives up a little resolution for a useful reduction in noise.	36 HU	0.44
`Cosine`	`cos(pi x / 2)`	Stronger roll-off than Shepp-Logan, intermediate between it and the smooth windows.	28 HU	0.36
`Hamming`	`0.54 + 0.46 cos(pi x)`	Smooth. Suppresses high frequencies heavily but retains a small non-zero response at the cutoff, so it preserves marginally more edge than Hann.	20 HU	0.25
`Hann`	`0.5 + 0.5 cos(pi x)`	The smoothest option: the window falls to exactly zero at the cutoff, giving the lowest noise and the softest edges. The natural choice for low-contrast detection in large patients, where noise, not resolution, is the limiting factor.	17.5 HU	0.22

Noise and MTF50 measured on the QA phantom at the default geometry. MTF50 in line pairs per millimetre. The two columns move together: this is the trade, and no kernel escapes it.



Ram-Lak **Shepp-Logan** **Hann**

The same low-dose QA acquisition reconstructed with three kernels. Noise falls from left to right; so does edge definition on the rods.

5.3 Unfiltered backprojection

Provided for contrast: the projections are backprojected with no filtering at all. The result is dominated by the one-over-radius blur inherent in the operation and is scaled so that the background region lands near its true level, since raw backprojection has an arbitrary offset. It is not a usable reconstruction; it is a demonstration of exactly how much work the ramp filter is doing.

5.4 SIRT

The simultaneous iterative reconstruction technique treats reconstruction as a least-squares problem and solves it by repeated forward and backward passes. At each iteration the current image estimate is projected, the residual against the measured sinogram is computed, and that residual is backprojected and added to the estimate. Row and column normalisation makes the step size independent of geometry.

$$x_{k+1} = x_k + C A^T R (p - A x_k) , R = 1 / \mathrm{rowsums}(A) , C = 1 / \mathrm{colsums}(A)$$

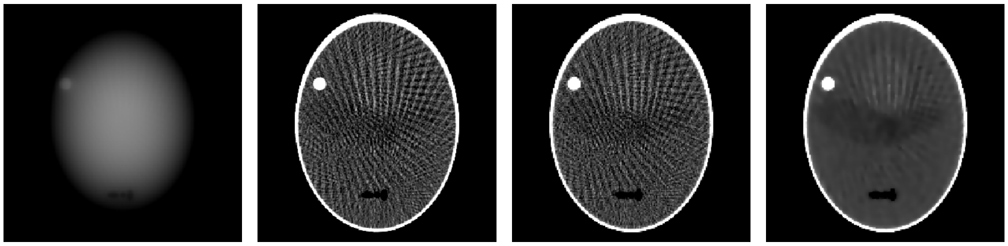
SIRT converges more slowly than filtered backprojection runs, but it handles incomplete and inconsistent data far more gracefully, because it never assumes the data are a complete Radon transform. With forty-five views it produces a usable image where filtered backprojection produces a starburst.

5.5 SIRT with total-variation regularisation

Between SIRT updates, two total-variation smoothing steps are applied. The step moves the image along the divergence of its normalised gradient, which smooths flat regions strongly while leaving strong edges nearly untouched, since the normalisation makes the flow independent of gradient magnitude.

$$u \leftarrow u + dt \times \mathrm{div} \left(\mathrm{grad}\, u / \mathrm{sqrt}(|\mathrm{grad}\, u|^2 + eps^2) \right)$$

The regularisation parameter eps sets the gradient below which a difference is treated as noise rather than structure. Inside the iteration the image is normalised so that one unit equals one thousand Hounsfield units, with eps fixed at 0.012, or twelve Hounsfield units. This is the closest thing here to a model-based iterative reconstruction, and it behaves the way commercial implementations do: excellent on sparse and photon-starved data, and capable of erasing real low-contrast structure if the weight is pushed too high.



Unfiltered BP **FBP** **SIRT, 15 it.** **SIRT + TV**

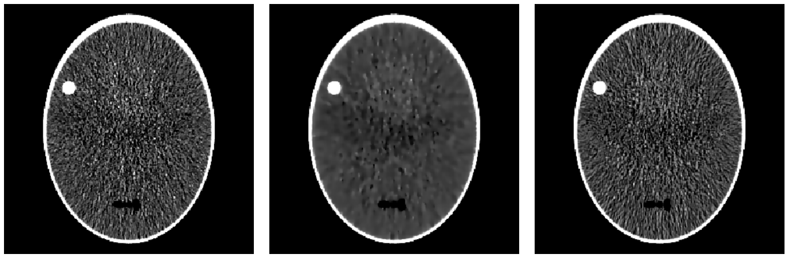
Head phantom from 45 projections. Filtered backprojection shows the classic sparse-view streaks; the iterative methods do not, because they never assume the missing angles were measured.

6. Denoising

Denoisers act on the reconstructed image in Hounsfield units, after reconstruction and before measurement. They are the cheapest way to make an image look better and the easiest way to destroy the information in it, so the metric panel is the honest referee: watch what happens to MTF50 and to contrast-to-noise ratio together, not to noise alone.

Filter	Internal parameters	Behaviour
`Gaussian`	`width = 0.8 × strength, in voxels`	Isotropic linear smoothing. Reduces noise efficiently and blurs edges by exactly the same amount, because it cannot tell them apart. Useful mainly as the baseline against which edge-preserving filters are judged.
`Median 3 by 3`	`one pass, two if strength > 1.6`	Rank filter. Excellent against isolated outliers such as single-channel spikes, leaves step edges largely intact, but erodes fine detail and can produce a blocky texture.
`Bilateral`	`spatial width 1.4 × strength voxels, range width 45 × strength HU`	Weights neighbours by both distance and intensity difference, so voxels across an edge contribute little. Preserves edges well at moderate strength. The range width sets what counts as an edge, so contrast below it is smoothed away.
`Total variation`	`step 3.0 × strength HU, 22 iterations, eps 12 HU`	The same flow used inside the iterative reconstruction, applied as a post-filter. Produces flat regions with sharp boundaries. At high strength it gives the characteristic cartoon or staircase appearance.
`Non-local means`	`bandwidth 28 × strength HU, search radius 5 voxels or 3 at large matrices, patch radius 1`	Averages voxels whose surrounding patches are similar, wherever they are in the search window. The strongest performer here on texture preservation, and by a wide margin the slowest.

None **Gaussian** **Median**



Bilateral **Total variation** **Non-local means**

The same low-dose head reconstruction under each denoiser at strength 1.2. All five reduce the visible noise; only some of them keep the skull edge and the small inserts.

The measured result on a low-dose head is worth stating plainly, because it is the point of having a metric panel at all. Gaussian smoothing drops MTF50 from 0.37 to 0.23 line pairs per millimetre. Bilateral, total-variation and non-local-means filtering achieve comparable noise reduction while leaving MTF50 essentially unchanged. An image that merely looks smoother is not evidence of anything; the resolution figure is what distinguishes filtering from discarding.

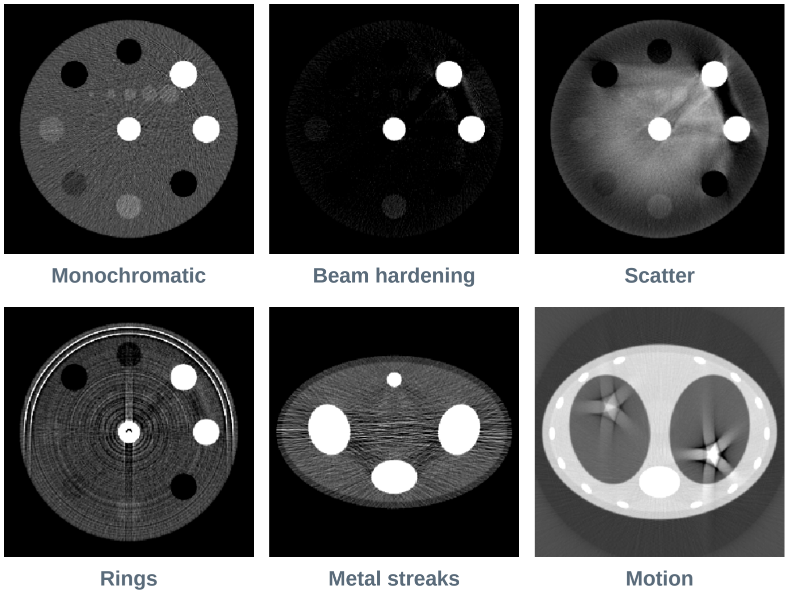
7. Image quality metrics

Twelve figures are computed on every run. Colour coding is fixed: green, amber and red correspond to the thresholds listed in the table.

Metric	Definition	Thresholds
`Voxel size`	Field of view divided by matrix size, in millimetres. The sampling limit on resolution, marked on the MTF plot as the voxel Nyquist frequency.	—
`Noise SD`	Standard deviation of Hounsfield values in the uniform background region.	green below 12 HU, amber below 35
`CNR`	Difference between the signal and background region means, divided by the background standard deviation. The figure that actually predicts whether a low-contrast object is detectable.	green above 5, amber above 2
`HU accuracy`	Reconstructed background mean minus its true value. Non-zero values indicate a systematic bias, most often beam hardening or uncorrected scatter.	green within 10 HU, amber within 40
`Cupping`	Central region mean minus peripheral region mean, both in the same nominally uniform material. The direct measure of beam hardening and scatter shading.	green within 10 HU, amber within 40
`RMSE`	Root mean square difference from ground truth over the object region, excluding surrounding air.	green below 60 HU, amber below 120
`PSNR`	Peak signal-to-noise ratio derived from the same masked error, in decibels.	—
`SSIM`	Structural similarity index. Computed after mapping Hounsfield values to a unit interval, since evaluating it directly on a scale where water is zero collapses the luminance term and yields a meaningless number.	green above 0.85, amber above 0.65
`MTF 50 and 10 per cent`	Spatial frequency at which modulation falls to half and to a tenth, in line pairs per millimetre. Measured by the circular-edge method: radial samples around a rod are binned at quarter-voxel spacing to form an oversampled edge profile, differentiated to a line spread function, tapered and transformed.	—
`Noise texture`	Mean frequency of the radial noise power spectrum. Low values indicate coarse, blotchy noise, which is the texture signature of over-regularised iterative reconstruction and is often more objectionable to a reader than higher but finer-grained noise.	—
`Compute`	Wall-clock time for the run.	—

The noise power spectrum requires a reasonably large uniform region and reports that the region is too small at the 64 and 128 matrices. This is by design rather than a fault: at those matrices the background region simply does not contain enough independent samples to estimate a spectrum.

8. Artifacts and how to recognise them



Each artifact isolated. The first two panels differ only in whether the beam is polychromatic.

Artifact	Appearance	Cause and control
`Beam hardening`	Depressed Hounsfield values toward the centre of large objects; dark bands between dense structures.	Preferential absorption of low-energy photons. Turn off the polychromatic beam to confirm: cupping should collapse to near zero.
`Scatter shading`	Broad, smooth loss of level and contrast, worst in the thickest regions.	Additive low-frequency signal before the logarithm. Controlled by the scatter to primary slider.
`Rings`	Concentric circles centred on the isocentre.	Per-channel gain errors that are stationary in the detector frame. Controlled by gain non-uniformity and dead channels.
`Metal streaks`	Bright and dark radiating bands between dense objects, with heavy local noise.	Photon starvation combined with severe beam hardening. Select the pelvis scenario; reduce with SIRT plus total variation.
`Motion`	Doubled or smeared edges with tangential streaks.	Inconsistent projections: the object is in different places in different views. Controlled by motion amplitude in the thorax scenario.
`Sparse-view streaks`	Fine straight lines radiating from high-contrast edges across the whole field.	Angular undersampling. Reduce the projection count to 45 or 90 to produce them; use an iterative method to suppress them.
`Photon starvation`	Locally extreme noise along the most attenuated ray paths, often with streaking.	Too few transmitted quanta for the logarithm to be meaningful. Lower the dose slider or select the pelvis scenario.

9. Guided experiments

Each experiment below changes one thing and states the numbers to expect, so a departure from them indicates either a genuinely different setting or a problem. Hold the seed fixed throughout, and use the pin button to keep the previous state on screen.

#	Procedure	Expected outcome
1	QA scenario. Step the kernel from Ram-Lak through to Hann, leaving everything else alone.	Noise falls from about 42 to 17.5 HU while MTF50 falls from 0.49 to 0.22 lp/mm. The two always move together.
2	Head scenario. Set detector elements to 128, then 256, then 512.	MTF50 rises through roughly 0.20, 0.36 and 0.56 lp/mm. Detector pitch, not matrix size, is setting the resolution.
3	Any scenario. Toggle the polychromatic beam off and back on.	Background moves from about -78 HU to +1.4 HU and cupping from -24 HU to +5.5 HU. That difference is the entire beam-hardening error.
4	Abdomen scenario at $10^{4.3}$ quanta. Switch between the two detector models.	Noise about 160 HU for the energy-integrating model against about 78 HU for photon counting. The gap closes at high dose.
5	Head scenario at 60 projections. Reconstruct with FBP, pin it, then switch to SIRT plus TV.	Noise falls from about 40.6 to 3.2 HU and SSIM rises from 0.72 to 0.96, with MTF50 held at 0.37 lp/mm. Resolution is not the price paid here.
6	Pelvis scenario. FBP, pin, then SIRT plus TV at weight 0.3.	Noise falls from about 76 to 48 HU, contrast-to-noise ratio rises from 0.37 to 0.80 and SSIM from 0.72 to 0.84.
7	Head scenario, dose lowered to $10^{4.2}$. Compare Gaussian against bilateral, TV and non-local means.	All reduce noise similarly. Only the Gaussian collapses MTF50, from 0.37 to 0.23 lp/mm.
8	QA scenario, monochromatic, high dose, Ram-Lak.	Water reconstructs at -0.2 HU with a standard deviation near 1.6 HU. This is the reference condition: if this is wrong, nothing else can be trusted.

9.1 Reference measurements

The following were measured during acceptance testing of the simulator itself and can be used to confirm a build is behaving.

Condition	Quantity	Value
Water cylinder, monochromatic, high dose	Reconstructed mu at centre	`0.19520 cm`$^{-1}$ `against a true 0.19500`
Same	Interior Hounsfield value	`-0.24 HU, SD 1.61 HU`
Head, default settings	Noise, CNR, SSIM, MTF50	`14.7 HU, 1.19, 0.929, 0.36 lp/mm`
Dose sweep across the full slider range	Noise	`9 HU to 889 HU`
Adjoint test of the iterative projector pair	Inner-product ratio	`1.000000`
Default head, 256 matrix	Compute time, FBP and SIRT+TV	`1.4 s and 4.6 s`

10. Limitations

The bench is a teaching and intuition instrument. It is quantitatively self-consistent, and the trends it produces are the right trends with roughly the right magnitudes, but it is not a substitute for a validated Monte Carlo simulation or for measurements on hardware. The following simplifications are deliberate and should be understood before any number from it is quoted.

Area	Simplification
`Geometry`	Two-dimensional parallel beam over 180 degrees. No fan or cone geometry, no helical pitch, no cone-beam artifacts, no z-direction sampling.
`Spectrum`	Five discrete energy bins approximating a filtered 120 kVp beam. No bowtie filter, no anode heel effect, no tube voltage control.
`Attenuation data`	Representative coefficients of the right order and with the right energy dependence, not a full NIST tabulation. Absolute Hounsfield values for bone, iodine and metal should be treated as indicative.
`Scatter`	An empirical smooth field scaled to a requested ratio, not a transport calculation. It reproduces the appearance and the Hounsfield bias, not the true spatial distribution.
`Detector physics`	Charge sharing is a one-dimensional three-tap redistribution. No pulse pileup, no depth-of-interaction effects, no explicit K-escape, no realistic pulse-height spectrum, no spectral binning on the photon-counting path.
`Corrections`	No beam-hardening correction, no scatter correction, no ring correction and no metal-artifact reduction are applied. What is displayed is the uncorrected physics, which is the point, but it means the images look worse than a modern scanner's output at equivalent settings.
`Motion`	Rigid periodic displacement of flagged objects. No deformable motion, no respiratory or cardiac model.
`Performance`	Reconstruction runs synchronously on the main thread. The 512 matrix with 720 projections will make the browser tab unresponsive for several seconds. This is a user-interface limitation, not a numerical one.

10.1 Reproducibility

All randomness derives from the seed control through a deterministic generator, so a given combination of settings and seed reproduces bit-for-bit on any machine. When comparing two configurations, hold the seed fixed so the difference is attributable to the parameter rather than to the noise draw. When judging whether a difference is real, vary the seed and see whether the difference survives.

CT Reconstruction Bench, manual revision 1.0. The application is a single self-contained HTML file with no external dependencies and no data collection of any kind.